

DATA ANALYSIS REPORT

Case study: customer shopping behaviour in a retail company

Author : Nguyễn Hà Phương Trang

E-mail : phuongtrangnguyenha.work@gmail.com

Tools for cleaning, structuring, and deep analysis:



Tools for visualization:



Ngày 20 tháng 06 năm 2026, TpHCM.

I.	EXECUTIVE SUMMARY	4
II.	DATASET DETAILS.....	5
III.	CLEAN & EXPLORATORY DATA ANALYSIS (PYTHON)	7
A.	Data sample	7
B.	Standardize format data	7
C.	Handling missing value	8
D.	Structuring	9
a)	Label:.....	9
b)	Segment:	9
E.	Grouping into age group.....	14
F.	Connect python to PostgreSQL.....	14
IV.	DEEP ANALYSIS (POSTGRESQL).....	15
A.	🎯 GOAL 1: REVENUE GROWTH	15
c)	Category & Purchased items:.....	15
d)	Seasonal patterns:.....	18
B.	🎯 GOAL 2: CUSTOMER RETENTION	20
a)	Customer Segmentation:	20
b)	Customer satisfaction – Review rating:	21
C.	🎯 GOAL 3: OPTIMIZE MARKETING CAMPAIGN.....	23
a)	Target customer:.....	23
b)	Payment method patterns:.....	25
V.	VISUALIZATION (POWER BI).....	26

Figure 1 Category's revenue ranking	15
Figure 2 Clothing & Accessories' revenue by season	15
Figure 3 Footwear & Outerwear' revenue by season.....	15
Figure 4 Purchase amount from discount using behaviour.....	16
Figure 5 % customer using discount	16
Figure 6 Using & Not using discount by category	16
Figure 7 Using & Not using discount by season.....	16
Figure 8 Top 3 highest revenue item by each category.....	17
Figure 9 Top 5 highest revenue of Size & Color of purchased item by each category	17
Figure 10 Top 5 lowest revenue of Size & Color of purchased item by each category	17
Figure 11 Average Purchase Amount by Season	18
Figure 12 number of New customers by each season	18
Figure 13 Revenue ranking by segment.....	18
Figure 14 revenue ranking by frequency of purchases.....	19
Figure 15 Member's total purchase amount	20
Figure 16 % customer are our member	20
Figure 17 age_group ranking by revenue	20
Figure 18 young adult's revenue by each category	20
Figure 19 Segment frequency	21
Figure 20 (%) bad review rating by category	21
Figure 21 (%) bad review by shipping type	22
Figure 22 (%) bad review by location & top 2 bad review shipping type	22
Figure 23 Young adult persona details.....	23
Figure 24 payment method details.....	25
Figure 25 Top 3 bad reviews of payment method by each age_group	25

I. EXECUTIVE SUMMARY

- **Business context:** This is a leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. The management team has noticed changes in purchasing patterns across demographics, product categories, and sales channels (online vs. offline). They are particularly interested in uncovering which factors, such as as discounts, reviews, seasons, or payment preferences, drive consumer decisions and repeat purchases.
- **Question:** "How can the company leverage consumer shopping data to identify trends, improve customer engagement, & optimize marketing and product strategies?"
- **Deliverables:**
 - **Identify patterns, trends, anomalies from** demographics, product categories, and sales channels.
 - **Investigate many factors affecting to consumer decisions and repeat purchases** such as discounts, reviews, seasons, or payment preferences.

II. DATASET DETAILS

- Dataset name: customer_shopping_behaviour
- Dataset details:
 - 3900 rows + 25 columns
 - Each row represent individual customer who has purchased items on the latest date.
- Data sample:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes

Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually

- Data type:

	customer_id	age	gender	item_purchased	category	purchase_amount_usd	location	size	color	season	review_rating
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3900.000000
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750538
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.713589
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000

subscription_status	shipping_type	discount_applied	promo_code_used	previous_purchases	payment_method	frequency_of_purchases	p
3900	3900	3900	3900	3900.000000	3900	3900	
2	6	2	2	NaN	6	7	
No	Free Shipping	No	No	NaN	PayPal	Every 3 Months	
2847	675	2223	2223	NaN	677	584	
NaN	NaN	NaN	NaN	25.351538	NaN	NaN	
NaN	NaN	NaN	NaN	14.447125	NaN	NaN	
NaN	NaN	NaN	NaN	1.000000	NaN	NaN	
NaN	NaN	NaN	NaN	13.000000	NaN	NaN	
NaN	NaN	NaN	NaN	25.000000	NaN	NaN	
NaN	NaN	NaN	NaN	38.000000	NaN	NaN	
NaN	NaN	NaN	NaN	50.000000	NaN	NaN	

ayment_method	frequency_of_purchases	purchase_label	r_score	f_score	m_score	review_score	rfm_score	segment	age_group
3900	3900	3900	2230.000000	3900.000000	3900.000000	3900.000000	2230.000000	3900	3900
6	7	2	NaN	NaN	NaN	NaN	NaN	10	4
PayPal	Every 3 Months	Others	NaN	NaN	NaN	NaN	NaN	Reconsider	Young adult
677	584	3817	NaN	NaN	NaN	NaN	NaN	1670	1028
NaN	NaN	NaN	2.968610	3.452821	2.980000	2.156667	2.996256	NaN	NaN
NaN	NaN	NaN	1.580756	1.152884	1.419503	0.751040	0.680606	NaN	NaN
NaN	NaN	NaN	1.000000	1.000000	1.000000	1.000000	1.000000	NaN	NaN
NaN	NaN	NaN	1.000000	2.000000	2.000000	2.000000	2.500000	NaN	NaN
NaN	NaN	NaN	2.000000	3.000000	3.000000	2.000000	2.950000	NaN	NaN
NaN	NaN	NaN	4.000000	4.000000	4.000000	3.000000	3.500000	NaN	NaN
NaN	NaN	NaN	5.000000	5.000000	5.000000	3.000000	4.700000	NaN	NaN

III. CLEAN & EXPLORATORY DATA ANALYSIS (PYTHON)

A. Data sample

```
[36]: import pandas as pd
```

```
[37]: df=pd.read_csv('customer_shopping_behavior.csv')
```

```
[38]: # DISCOVERY & CLEAN
```

```
[39]: df.head()
```

```
[39]:
```

	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually

B. Standardize format data

```
[40]: df.dtypes
```

```
[40]: Customer ID      int64
Age                int64
Gender             object
Item Purchased     object
Category           object
Purchase Amount (USD)  int64
Location           object
Size              object
Color             object
Season            object
Review Rating      float64
Subscription Status object
Shipping Type      object
Discount Applied   object
Promo Code Used    object
Previous Purchases int64
Payment Method     object
Frequency of Purchases object
dtype: object
```

```
[41]: #snake_casing format
```

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
```

```
[42]: df.columns
```

```
[42]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount_(usd)', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='object')
```

C. Handling missing value

```
[43]: df = df.rename(columns={'purchase_amount_(usd)' : 'purchase_amount_usd' })
```

```
[44]: df.columns
```

```
[44]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',  
        'purchase_amount_usd', 'location', 'size', 'color', 'season',  
        'review_rating', 'subscription_status', 'shipping_type',  
        'discount_applied', 'promo_code_used', 'previous_purchases',  
        'payment_method', 'frequency_of_purchases'],  
        dtype='object')
```

```
[45]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 3900 entries, 0 to 3899  
Data columns (total 18 columns):  
#   Column                Non-Null Count  Dtype  ---  
0   customer_id           3900 non-null   int64  
1   age                   3900 non-null   int64  
2   gender                3900 non-null   object  
3   item_purchased        3900 non-null   object  
4   category              3900 non-null   object  
5   purchase_amount_usd   3900 non-null   int64  
6   location              3900 non-null   object  
7   size                  3900 non-null   object  
8   color                 3900 non-null   object  
9   season                3900 non-null   object  
10  review_rating          3863 non-null   float64  
11  subscription_status    3900 non-null   object  
12  shipping_type          3900 non-null   object  
13  discount_applied       3900 non-null   object  
14  promo_code_used        3900 non-null   object  
15  previous_purchases     3900 non-null   int64  
16  payment_method         3900 non-null   object  
17  frequency_of_purchases 3900 non-null   object
```

```
[46]: df['review_rating'] = df.groupby("category")['review_rating'].transform(lambda x : x.fillna(df['review_rating'].median()))
```

```
[47]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 3900 entries, 0 to 3899  
Data columns (total 18 columns):  
#   Column                Non-Null Count  Dtype  ---  
0   customer_id           3900 non-null   int64  
1   age                   3900 non-null   int64  
2   gender                3900 non-null   object  
3   item_purchased        3900 non-null   object  
4   category              3900 non-null   object  
5   purchase_amount_usd   3900 non-null   int64  
6   location              3900 non-null   object  
7   size                  3900 non-null   object  
8   color                 3900 non-null   object  
9   season                3900 non-null   object  
10  review_rating          3900 non-null   float64  
11  subscription_status    3900 non-null   object  
12  shipping_type          3900 non-null   object  
13  discount_applied       3900 non-null   object  
14  promo_code_used        3900 non-null   object  
15  previous_purchases     3900 non-null   int64  
16  payment_method         3900 non-null   object  
17  frequency_of_purchases 3900 non-null   object  
dtypes: float64(1), int64(4), object(13)  
memory usage: 548.6+ KB
```


D. Structuring

a) Label:

```
[49]: #Label Second Buy vs Others
#=====
import numpy as np

df['purchase_label'] = np.where(
    df['previous_purchases'] < 2,
    'Second Buy', # Điều kiện True
    'Others'      # Điều kiện False
)

print(df[['previous_purchases', 'purchase_label']].head(8))
```

	previous_purchases	purchase_label
0	14	Others
1	2	Others
2	23	Others
3	49	Others
4	31	Others
5	14	Others
6	49	Others
7	19	Others

b) Segment:

```
# Segmentation following:
# 1. r (frequency of purchase) 2. f (previous purchases) 3. m (purchase amount) 4. rv (review rate)
# =====
# BƯỚC 1: RECENCY SCORE — từ frequency_of_purchases
# =====

recency_map = {
    'Bi-weekly': 6,
    'Fortnightly': 5,
    'Monthly': 4,
    'Every 3 months': 3,
    'Quarterly': 2,
    'Annually': 1
}

df['r_score'] = df['frequency_of_purchases'].map(recency_map)

# =====
# BƯỚC 2: FREQUENCY SCORE — từ previous_purchase
# =====

df['f_score'] = np.where(
    df['previous_purchases'] < 2,
    1, # Second Buy → F thấp nhất
    pd.qcut(
        df['previous_purchases'].clip(lower=2),
        q=4,
        labels=[2, 3, 4, 5],
        duplicates='drop'
    ).astype(float)
)
```

```

# =====
# BƯỚC 3: MONETARY SCORE — từ purchase_amount
# =====

df['m_score'] = pd.qcut(
    df['purchase_amount_usd'],
    q=5,
    labels=[1, 2, 3, 4, 5],
    duplicates='drop'
).astype(float)

# =====
# BƯỚC 4: REVIEW SCORE — từ review_rate
# =====

df['review_score'] = pd.cut(
    df['review_rating'],
    bins=[0, 3, 4, 5],
    labels=[1, 2, 3] # 1-3 → Thấp | 3-4 → Trung bình | 4-5 → Cao
).astype(float)

# =====
# BƯỚC 5: TỔNG HỢP RFM + REVIEW
# =====

df['rfm_score'] = (
    df['r_score'] * 0.25 +
    df['f_score'] * 0.30 +
    df['m_score'] * 0.30 +
    df['review_score'] * 0.15
).round(2)

# =====
# BƯỚC 6: GÁN SEGMENT
# =====

def assign_segment(row):
    r = row['r_score']
    f = row['f_score']
    m = row['m_score']
    rv = row['review_score']

    # _____
    # GUARD CLAUSE — Kiểm tra null trước
    # Bất kỳ score nào null → không đủ data
    # → Xếp vào Reconsider, không tính RFM
    # _____
    if pd.isna(r) or pd.isna(f) or pd.isna(m) or pd.isna(rv):

```

```
return 'Reconsider'
```

```
# _____
```

```
# 1. CHAMPIONS — Khách hàng VIP
```

```
# R, F, M đều cao nhất + hài lòng cao
```

```
# _____
```

```
if r >= 4 and f >= 4 and m >= 4 and rv >= 3:
```

```
    return 'Champions'
```

```
# _____
```

```
# 2. LOYAL CUSTOMERS — Khách hàng trung thành
```

```
# F và M cao, quay lại thường xuyên + hài lòng
```

```
# Không nhất thiết vừa mua gần đây
```

```
# _____
```

```
elif f >= 4 and m >= 3 and rv >= 3:
```

```
    return 'Loyal Customers'
```

```
# _____
```

```
# 3. POTENTIAL LOYALIST — Khách hàng tiềm năng
```

```
# Mới mua gần đây, tần suất bắt đầu đều
```

```
# Chi tiêu khá, review trung bình trở lên
```

```
# _____
```

```
elif r >= 4 and f >= 2 and m >= 3 and rv >= 2:
```

```
    return 'Potential Loyalist'
```

```
# _____
```

```
# 4. NEW CUSTOMERS — Khách hàng mới
```

```
# R rất cao (vừa mua gần đây)
```

```
# F và M còn thấp, review chưa có nhiều
```

```
# → Không filter rv vì khách mới chưa đủ data đánh giá
```

```
# _____
```

```
elif r >= 4 and f <= 2 and m <= 2:
```

```
    return 'New Customers'
```

```
# _____
```

```
# 5. PROMISING — Khách hàng hứa hẹn
```

```
# Tần suất ổn, đang tăng dần
```

```
# Giá trị chưa cao nhưng review tích cực
```

```
# _____
```

```
elif r >= 3 and f >= 3 and m <= 3 and rv >= 2:
```

```
    return 'Promising'
```

```
# _____
```

```
# 6. NEEDS ATTENTION — Khách cần quan tâm
```

```
# Tất cả chỉ số ở mức trung bình
```

```
# rv trung bình → chưa đủ hài lòng để gắn bó lâu dài
```

```
# _____
```

```
elif r >= 2 and f >= 2 and m >= 2 and rv <= 1:
```

```

return 'Needs Attention'

# -----
# 7. CAN'T LOSE THEM — Không thể mất
# Từng mua rất nhiều & thường xuyên (F, M cao)
# R thấp — đang rời đi
# rv thấp → không hài lòng là lý do rời đi
# -----
elif r <= 2 and f >= 4 and m >= 4 and rv <= 1:
    return "Can't Lose Them"

# -----
# 8. AT RISK — Nguy cơ mất khách
# Từng mua khá nhưng lâu rồi chưa quay lại
# rv thấp → có thể là lý do giảm tương tác
# -----
elif r <= 2 and (f >= 3 or m >= 3) and rv <= 1:
    return 'At Risk'

# -----
# 9. HIBERNATING / LOST — Khách rời bỏ
# R, F, M đều thấp + review tệ
# Lâu không tương tác, gần như mất hẳn
# -----
else:
    return 'Hibernating/Lost'

df['segment'] = df.apply(assign_segment, axis=1)

# =====
# BƯỚC 7: KIỂM TRA KẾT QUẢ
# =====

# Phân phối segment
print("=== PHÂN PHỐI SEGMENT ===")
print(df['segment'].value_counts())
print(f"\nTổng khách: {len(df)}")

# Profile trung bình từng segment
print("\n=== PROFILE TRUNG BÌNH TỪNG SEGMENT ===")
profile = df.groupby('segment').agg(
    so_luong_khach = ('segment', 'count'),
    r_score_tb = ('r_score', 'mean'),
    f_score_tb = ('f_score', 'mean'),
    m_score_tb = ('m_score', 'mean'),
    review_rating_tb = ('review_rating', 'mean'),
    rfm_score_tb = ('rfm_score', 'mean'),

```

```

purchase_amt_tb = ('purchase_amount_usd', 'mean'),
prev_purchase_tb = ('previous_purchases', 'mean')
).round(2)

print(profile)

```

```

=== PHÂN PHỐI SEGMENT ===
segment
Reconsiderer      1670
Hibernating/Lost  836
Potential Loyalist 373
Needs Attention   260
Promising         257
Loyal Customers   181
New Customers     115
At Risk           112
Champions         80
Can't Lose Them   16
Name: count, dtype: int64

Tổng khách: 3900

=== PROFILE TRUNG BÌNH TỪNG SEGMENT ===
              so_luong_khach  r_score_tb  f_score_tb  m_score_tb \
segment
At Risk                112         1.23         3.58         2.29
Can't Lose Them         16         1.00         4.50         4.56
Champions                80         4.47         4.46         4.49
Hibernating/Lost       836         1.69         3.28         2.77
Loyal Customers         181         2.28         4.46         3.77
Needs Attention         260         3.62         3.52         3.52
New Customers           115         4.59         1.91         1.50
Potential Loyalist      373         4.47         3.19         3.95
Promising               257         4.47         3.96         1.48
Reconsiderer           1670         NaN         3.44         2.99

              review_rating_tb  rfm_score_tb  purchase_amt_tb \
segment
At Risk                2.77         2.22         49.07
Can't Lose Them         2.79         3.12         86.06
Champions                4.54         4.25         83.50
Hibernating/Lost        3.82         2.57         56.06
Loyal Customers          4.56         3.49         72.91
Needs Attention          2.77         3.17         68.70
New Customers            3.74         2.49         35.64
Potential Loyalist       3.89         3.61         75.82
Promising                4.02         3.12         35.61
Reconsiderer             3.75         NaN         59.92

              prev_purchase_tb
segment
At Risk                27.01
Can't Lose Them        38.81
Champions               37.64
Hibernating/Lost        23.24
Loyal Customers         37.58
Needs Attention         26.25

```

E. Grouping into age group

```
[52]: age_labels = ["Young adult", "adult", "Middle-aged", "Senior"]

# Tạo cột mới 'Age_Group' bằng pd.cut
df['age_group'] = pd.cut(df['age'], bins=4, labels=age_labels)

[53]: df.columns

[53]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
        'purchase_amount_usd', 'location', 'size', 'color', 'season',
        'review_rating', 'subscription_status', 'shipping_type',
        'discount_applied', 'promo_code_used', 'previous_purchases',
        'payment_method', 'frequency_of_purchases', 'purchase_label', 'r_score',
        'f_score', 'm_score', 'review_score', 'rfm_score', 'segment',
        'age_group'],
        dtype='object')
```

F. Connect python to PostgreSQL

```
[55]: !pip install sqlalchemy psycopg2

Requirement already satisfied: sqlalchemy in c:\users\admin\appdata\local\programs\python\python313\lib\site-packages (2.0.50)
Requirement already satisfied: psycopg2 in c:\users\admin\appdata\local\programs\python\python313\lib\site-packages (2.9.12)
Requirement already satisfied: greenlet>=1 in c:\users\admin\appdata\local\programs\python\python313\lib\site-packages (from sqlalchemy) (3.5.1)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\users\admin\appdata\local\programs\python\python313\lib\site-packages (from sqlalchemy) (4.12.2)

[notice] A new release of pip is available: 25.3 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip

[56]: from sqlalchemy import create_engine

username = 'postgres'
password = 'Highlands'
host = 'localhost'
port = '5432'
database = 'Customer_Shopping_Behavior'

engine = create_engine('postgresql+psycopg2://postgres:Highlands@localhost:5432/Customer_Shopping_Behavior')

# 3. Đẩy toàn bộ dữ liệu vào Database chỉ với 1 dòng lệnh!
try:
    # if_exists='replace': Nếu bảng đã có, nó sẽ xóa đi và tạo lại bản chuẩn xác nhất
    # index=False: Không đưa cột số thứ tự mặc định của Python vào Database
    df.to_sql('shopping_event', engine, if_exists='replace', index=False)
    print("👉 Import dữ liệu thành công rực rỡ!")
except Exception as e:
    print("Có lỗi xảy ra:", e)

👉 Import dữ liệu thành công rực rỡ!
```

IV. DEEP ANALYSIS (POSTGRESQL)

A. 🎯 GOAL 1: REVENUE GROWTH

c) Category & Purchased items:

Question: "Which product is generating the most revenue for me, and how should I run seasonal campaigns for that product?"

1. Which category has the highest purchase amount? By season?

Clothing category has the highest revenue compared to Accessories, Footwear, and Outerwear.

	category text	sum numeric
1	Clothing	104264
2	Accessories	74200
3	Footwear	36093
4	Outerwear	18524

Figure 1 Category's revenue ranking

The following figures are the categories' revenue by each season:

	season text	category text	sum numeric		season text	category text	sum numeric
1	Spring	Clothing	27692	1	Fall	Accessories	19874
2	Winter	Clothing	27274	2	Summer	Accessories	19028
3	Fall	Clothing	26220	3	Winter	Accessories	18291
4	Summer	Clothing	23078	4	Spring	Accessories	17007

Figure 2 Clothing & Accessories' revenue by season

	season text	category text	sum numeric		season text	category text	sum numeric
1	Spring	Footwear	9555	1	Fall	Outerwear	5259
2	Summer	Footwear	9393	2	Winter	Outerwear	4562
3	Fall	Footwear	8665	3	Spring	Outerwear	4425
4	Winter	Footwear	8480	4	Summer	Outerwear	4278

Figure 3 Footwear & Outerwear' revenue by season

2. Does discount using behaviour boost our sales?

The below table indicates our total revenue comes from 42.65% which is used discount vouchers. So, this is normal when we have to decrease product's price after applying vouchers.

	%Purchase amount Using discount numeric	%Purchase amount No Discount numeric
1	42.65	57.35

Figure 4 Purchase amount from discount using behaviour

Additionally, 43% of our total customer are using discounts.

	%Using Discount numeric
1	43.00

Figure 5 % customer using discount

Our sales does not dependent on discount, this is a positive sign for organic revenue stream.

	category text	%Using Discount numeric		category text	%Not Using Discount numeric
1	Clothing	18.74	1	Clothing	25.79
2	Accessories	13.92	2	Accessories	17.87
3	Footwear	6.64	3	Footwear	8.72
4	Outerwear	3.69	4	Outerwear	4.62

Figure 6 Using & Not using discount by category

	season text	%Using Discount numeric		season text	%Not Using Discount numeric
1	Spring	11.28	1	Fall	14.82
2	Summer	10.85	2	Spring	14.33
3	Winter	10.69	3	Winter	14.21
4	Fall	10.18	4	Summer	13.64

Figure 7 Using & Not using discount by season

3. Top 3 highest revenue of purchased Items by each Category.

	category text	item_purchased text	total numeric	rank_ bigint
1	Accessories	Jewelry	10010	1
2	Accessories	Sunglasses	9649	2
3	Accessories	Belt	9635	3
4	Clothing	Blouse	10410	1
5	Clothing	Shirt	10332	2
6	Clothing	Dress	10320	3
7	Footwear	Shoes	9240	1
8	Footwear	Sandals	9200	2
9	Footwear	Boots	9018	3
10	Outerwear	Coat	9275	1
11	Outerwear	Jacket	9249	2

Figure 8 Top 3 highest revenue item by each category

4. Top 5 highest/ lowest revenue of Size & Color of purchased item by each category.

	category text	size text	color text	item_purchased text	total numeric	rank_ bigint
1	Accessories	M	Gray	Jewelry	490	1
2	Accessories	M	Black	Backpack	455	2
3	Accessories	M	Blue	Belt	453	3
4	Clothing	M	Green	Hoodie	541	1
5	Clothing	M	Teal	Pants	508	2
6	Clothing	L	Yellow	Shorts	485	3
7	Footwear	M	Violet	Boots	527	1
8	Footwear	M	Purple	Sandals	459	2
9	Footwear	M	Gold	Shoes	412	3
10	Outerwear	M	Violet	Coat	484	1
11	Outerwear	M	Blue	Jacket	410	2
12	Outerwear	M	Gray	Jacket	374	3

Figure 9 Top 5 highest revenue of Size & Color of purchased item by each category

	category text	size text	color text	item_purchased text	total numeric	rank_ bigint
1	Accessories	XL	Gold	Scarf	20	1
2	Accessories	XL	Teal	Belt	20	2
3	Accessories	S	Blue	Hat	20	3
4	Clothing	XL	Red	Hoodie	20	1
5	Clothing	XL	Red	Skirt	20	2
6	Clothing	S	Purple	Pants	20	3
7	Footwear	M	Lavend...	Sneakers	20	1
8	Footwear	XL	Beige	Sandals	20	2
9	Footwear	M	White	Sneakers	20	3
10	Outerwear	L	Indigo	Jacket	21	1
11	Outerwear	XL	Black	Coat	22	2
12	Outerwear	S	Orange	Coat	23	3

Figure 10 Top 5 lowest revenue of Size & Color of purchased item by each category

d) Seasonal patterns:

Question: "When are customers most willing to spend money?"

1. Average Purchase Amount by Season — Which season generates the highest revenue?

	season text	avg_purchase_amount numeric
1	Fall	61.56
2	Winter	60.36
3	Spring	58.74
4	Summer	58.41

Figure 11 Average Purchase Amount by Season

2. Which season has the highest number of new customers?

R is very high (recently purchased)

F and M are still low, not many reviews yet

→ No filter reviews because new customers don't have enough data to leave a review

	season text	segment text	count bigint
1	Fall	New Customers	33
2	Winter	New Customers	29
3	Summer	New Customers	27
4	Spring	New Customers	26

Figure 12 number of New customers by each season

3. Which segment generates higher overall revenue?

	segment text	sum numeric
1	Reconsider	100074
2	Hibernating/Lost	46864
3	Potential Loyalist	28282
4	Needs Attention	17861
5	Loyal Customers	13196
6	Promising	9152
7	Champions	6680
8	At Risk	5496
9	New Customers	4099
10	Can't Lose Them	1377

Figure 13 Revenue ranking by segment

4. How effective are customers who buy based on frequency?

	frequency_of_purchases text	sum numeric
1	Every 3 Months	35088
2	Annually	34419
3	Quarterly	33771
4	Bi-Weekly	33200
5	Monthly	32810
6	Fortnightly	32007
7	Weekly	31786

Figure 14 revenue ranking by frequency of purchases

B. 🎯 GOAL 2: CUSTOMER RETENTION

a) Customer Segmentation:

Segmentation (Frequency of Purchases, Previous purchase, Purchase amount, Review rate)

Question: "Who is at risk of leaving without my knowledge?"

1. Do customers with subscriptions (yes) have higher purchase amounts and frequencies than non-subscribers? → This is proof to push subscriptions.

	Member's purchase amount % numeric
1	26.88

Figure 15 Member's total purchase amount

	Member's retail % numeric
1	27.00

Figure 16 % customer are our member

2. Customer group Frequency = Annually or Every 3 months — what do they buy, where do they buy (Location), and how old are they?

→ This is the at-risk churn group [Young adult], who brings high value order but intends to leave us.

	age_group text	sum numeric
1	Young adult	18440
2	adult	17464
3	Senior	17065
4	Middle-aged	16538

Figure 17 age_group ranking by revenue

	category text	sum numeric
1	Clothing	9413
2	Accessories	4991
3	Footwear	2464
4	Outerwear	1572

Figure 18 young adult's revenue by each category

3. High Previous Purchases (multiple previous purchases) but low Review Rate (1-2) / Low Frequency → Customers are about to leave, even if they were once loyal. How many people are like that? [Can't Lose them & At risk]

	segment text	%NO_Customers numeric
1	Reconsider	42.82
2	Hibernating/Lost	21.44
3	Potential Loyalist	9.56
4	Needs Attention	6.67
5	Promising	6.59
6	Loyal Customers	4.64
7	New Customers	2.95
8	At Risk	2.87
9	Champions	2.05
10	Can't Lose Them	0.41

Figure 19 Segment frequency

- b) Customer satisfaction – Review rating:

Question: "Where are the low reviews in my store?"

1. Which category has the worst review rate?

	category text	bad_review% numeric
1	Outerwear	19.1
2	Clothing	18.5
3	Accessories	16.2
4	Footwear	16.0

Figure 20 (%) bad review rating by category

2. Customer review rating = < 3 star — Which shipping method did they use?
(Delivery issues?)

	shipping_type text	%bad_review numeric
1	Store Pickup	21.23
2	Next Day Air	18.83
3	Express	16.72
4	2-Day Shipping	16.27
5	Free Shipping	16.30
6	Standard	15.29

Figure 21 (%) bad review by shipping type

3. Where are low review rates most common?

→ This could be a regional/warehouse issue.

	location text	%bad_reviews numeric
1	Oklahoma	28.0
2	West Virginia	25.9
3	Connecticut	25.6
4	Georgia	24.1
5	Arizona	23.1
6	South Dakota	22.9
7	Wyoming	22.5
8	New Hamp...	22.5
9	Michigan	21.9
10	Illinois	21.7

	location text	shipping_type text	bad_review_pct numeric	rn bigint
1	Arizona	Next Day Air	36.4	1
2	Arizona	Store Pickup	33.3	2
3	Arkansas	Free Shipping	35.7	1
4	Arkansas	2-Day Shipping	33.3	2
5	Connecticut	Express	54.5	1
6	Connecticut	2-Day Shipping	31.3	2
7	Georgia	2-Day Shipping	38.5	1
8	Georgia	Next Day Air	33.3	2
9	Michigan	Store Pickup	36.4	1
10	Michigan	Free Shipping	33.3	2
11	New Hamp...	Next Day Air	36.4	1
12	New Hamp...	Free Shipping	27.3	2
13	Oklahoma	Free Shipping	37.5	1
14	Oklahoma	Next Day Air	35.7	2
15	South Dak...	Express	35.7	1
16	South Dak...	Free Shipping	28.6	2
17	West Virgi...	2-Day Shipping	33.3	1
18	West Virgi...	Next Day Air	28.6	2
19	Wyoming	Standard	36.4	1
20	Wyoming	Free Shipping	30.0	2

Figure 22 (%) bad review by location & top 2 bad review shipping type

C. 🎯 GOAL 3: OPTIMIZE MARKETING CAMPAIGN

a) Target customer:

Question: "Who am I talking to, and am I speaking their language correctly?"

1. Which age group spends the most?

	age_group text	total_spend numeric		segment text	%NO_customers numeric
1	Young adult	62143	1	Reconsider	44.7
2	Middle-aged	59197	2	Hibernating/Lost	19.6
3	adult	55978	3	Potential Loyalist	9.5
4	Senior	55763	4	Needs Attention	6.7
			5	Loyal Customers	6.1
			6	Promising	5.4
			7	New Customers	3.1
			8	At Risk	2.4
			9	Champions	2.0
			10	Can't Lose Them	0.3

Figure 23 Young adult persona details

2. Which customers pay more than avg after they already applied vouchers.

	customer_id bigint	segment text	age_group text
1	582	Reconsider	adult
2	1422	Needs Attention	Senior
3	1209	Hibernating/Lo...	Young adult
4	2072	Loyal Custome...	Senior
5	2808	Potential Loyal...	Middle-aged
6	2843	Loyal Custome...	Young adult
7	3029	Reconsider	Senior
8	1457	Reconsider	Senior
9	3838	Champions	Middle-aged
10	519	Reconsider	Young adult
11	1848	Reconsider	Middle-aged
12	2972	Reconsider	Young adult
13	194	Can't Lose Them	adult
14	205	Reconsider	Young adult
15	3266	Potential Loyal...	adult
16	1480	Reconsider	Middle-aged
Total rows: 1963		Query complete 00:00:00.207	

Figure 24 customers pay more than avg after they already applied vouchers

2.1. How about Segment & age_group distribution from these customers?

	age_group text	no_customers bigint		segment text	no_customers bigint
			1	Reconsider	371
			2	Potential Loyalist	144
			3	Hibernating/Lost	140
			4	Needs Attention	74
1	Young adult	229	5	Loyal Customers	58
2	Senior	204	6	Champions	27
3	Middle-aged	204	7	At Risk	17
4	adult	202	8	Can't Lose Them	8

Figure 25 age_group & segment distribution pay more than avg after they already applied vouchers

	age_group text	segment text	no_customers bigint
1	Middle-aged	Reconsider	103
2	Young adult	Reconsider	102
3	Senior	Reconsider	85
4	adult	Reconsider	81
5	adult	Potential Loyalist	42
6	adult	Hibernating/Lost	40
7	Young adult	Potential Loyalist	39
8	Senior	Hibernating/Lost	38
9	Middle-aged	Potential Loyalist	33
10	Young adult	Hibernating/Lost	31
11	Middle-aged	Hibernating/Lost	31
12	Senior	Potential Loyalist	30
Total rows: 32		Query complete 00:00:00.344	

b) Payment method patterns:

1. Which payment method is most popular according to age group? → Design a campaign that suits the payment channel.

	age_group text	payment_method text	frequency bigint	rank_ bigint
1	adult	Venmo	166	1
2	adult	Debit Card	163	2
3	adult	Credit Card	161	3
4	Middle-aged	PayPal	193	1
5	Middle-aged	Cash	172	2
6	Middle-aged	Debit Card	161	3
7	Senior	Credit Card	177	1
8	Senior	Cash	166	2
9	Senior	Venmo	155	3
10	Young adult	Cash	181	1
11	Young adult	Credit Card	177	2
12	Young adult	PayPal	172	3

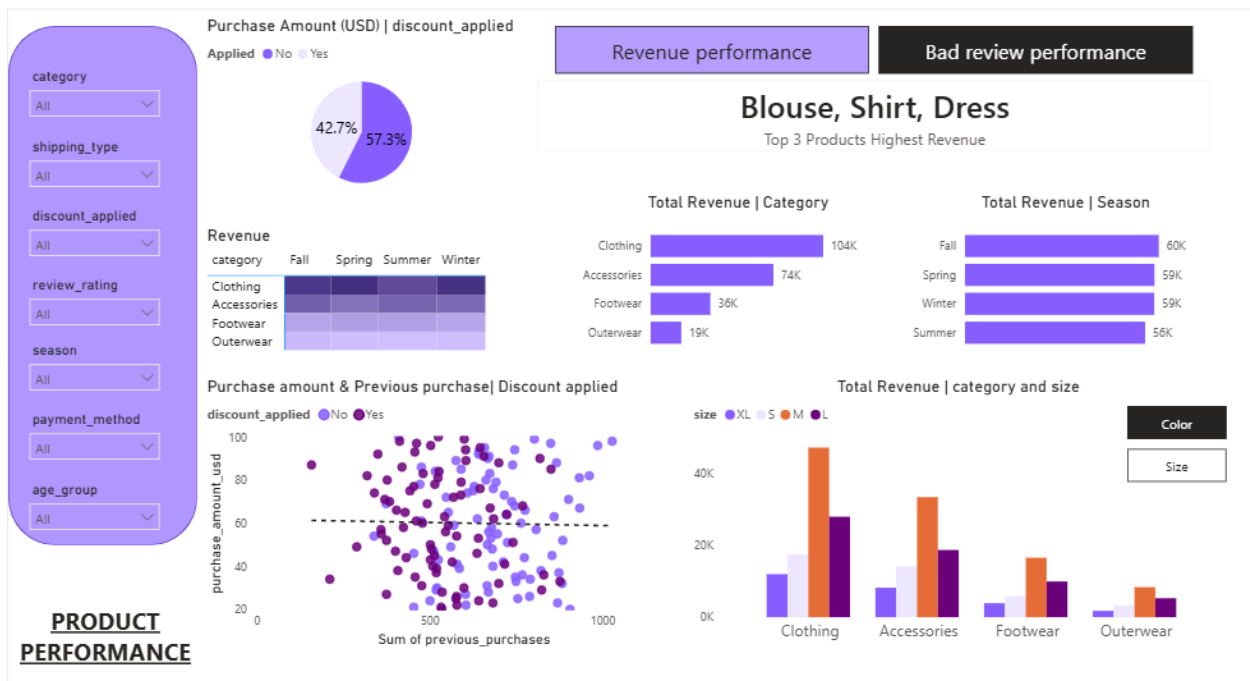
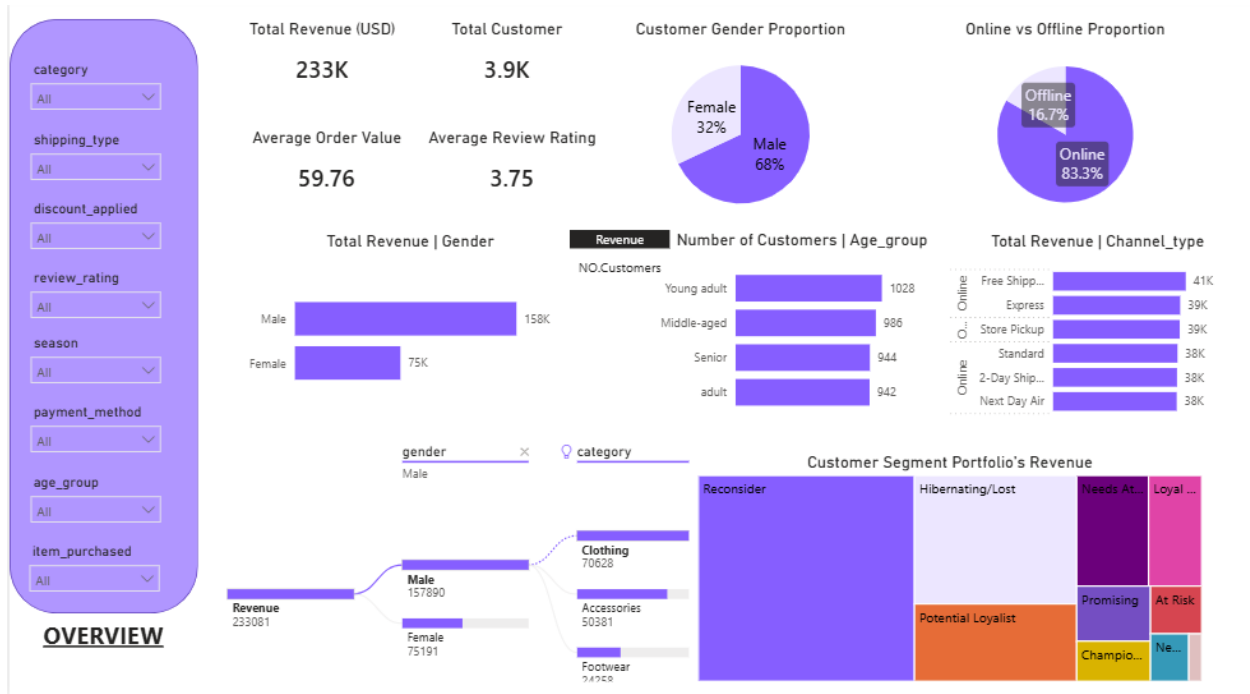
	payment_method text	frequency bigint
1	PayPal	677
2	Credit Card	671
3	Cash	670
4	Debit Card	636
5	Venmo	634
6	Bank Transfer	612

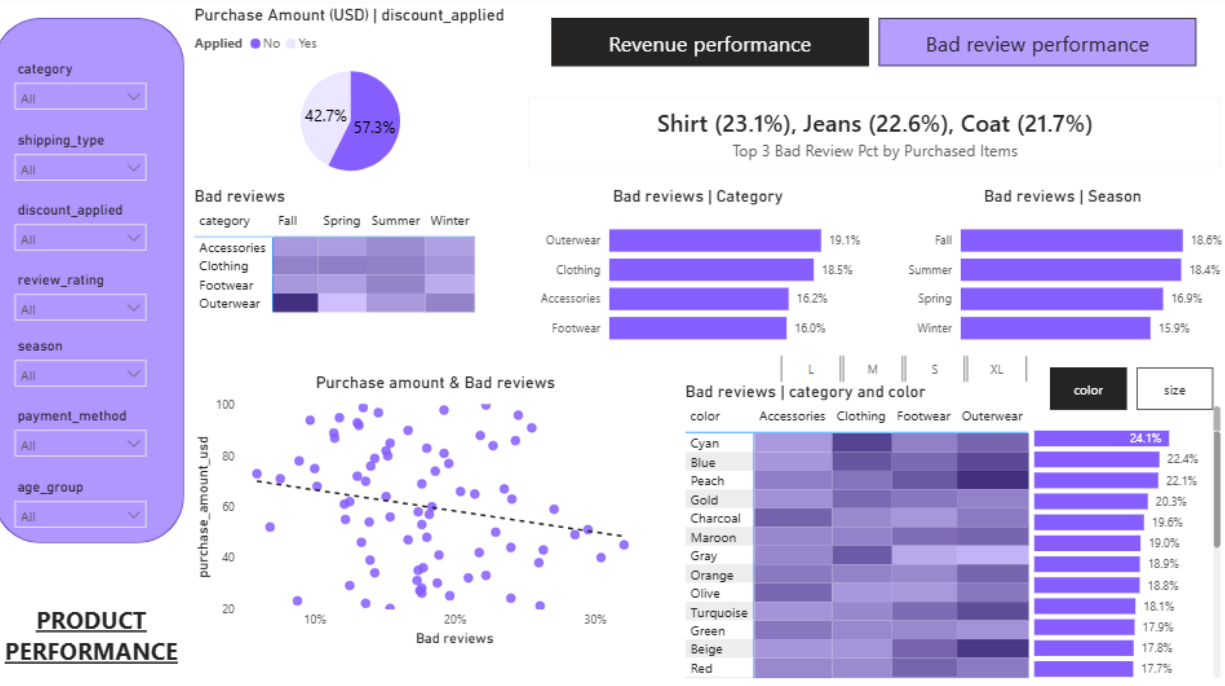
Figure 26 payment method details

	age_group text	payment_method text	%bad_reviews numeric	rank_ bigint
1	adult	Bank Transfer	22.1	1
2	adult	Cash	19.9	2
3	adult	Debit Card	19.6	3
4	Middle-aged	Bank Transfer	23.7	1
5	Middle-aged	Cash	22.1	2
6	Middle-aged	Venmo	20.3	3
7	Senior	Cash	18.7	2
8	Senior	Credit Card	18.1	3
9	Senior	PayPal	20.5	1
10	Young adult	Bank Transfer	15.9	2
11	Young adult	Debit Card	19.6	1
12	Young adult	Venmo	15.8	3

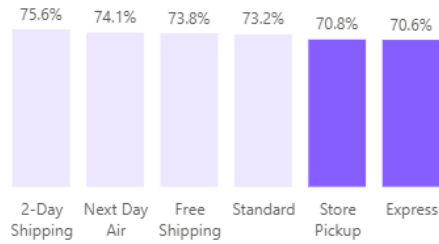
Figure 27 Top 3 bad reviews of payment method by each age_group

V. VISUALIZATION (POWER BI)

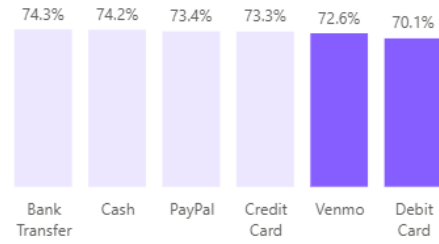




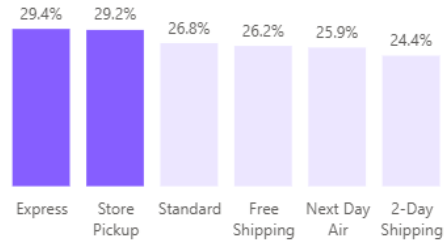
% Non_Member by shipping_type



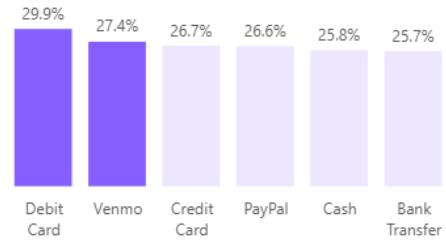
% Non_Member by payment_method



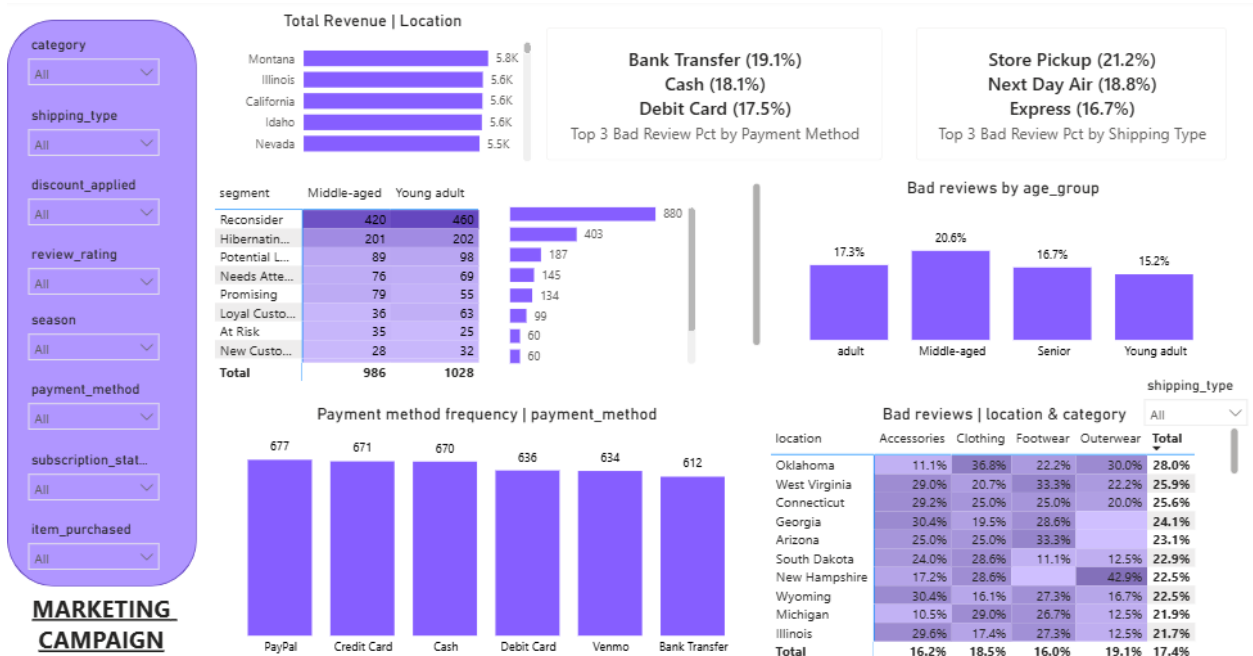
% Member by shipping_type



% Member by payment_method



CUSTOMER RETENTION



MARKETING CAMPAIGN